

## SAFETY DATA SHEET

Creation Date 16-Jan-2009

Revision Date 26-Mar-2024

Revision Number 3

### 1. Identification

**Product Name** (S)-(-)-Glycidol

**Cat No. :** H63502

**CAS No** 60456-23-7  
**Synonyms** (S)-Oxiranemethanol

**Recommended Use** Laboratory chemicals.  
**Uses advised against** Food, drug, pesticide or biocidal product use.

#### Details of the supplier of the safety data sheet

##### Company

Thermo Fisher Scientific Chemicals, Inc.  
30 Bond Street  
Ward Hill, MA 01835-8099  
Tel: 800-343-0660  
Fax: 800-322-4757

##### **Emergency Telephone Number**

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11  
Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99  
**CHEMTREC** Tel. No. **US**:001-800-424-9300 / **Europe**:001-703-527-3887

### 2. Hazard(s) identification

#### **Classification**

This chemical is considered hazardous by the 2012 OSHA Hazard Communication Standard (29 CFR 1910.1200)

|                                                  |             |
|--------------------------------------------------|-------------|
| Flammable liquids                                | Category 4  |
| Acute oral toxicity                              | Category 4  |
| Acute dermal toxicity                            | Category 4  |
| Acute Inhalation Toxicity - Vapors               | Category 3  |
| Skin Corrosion/Irritation                        | Category 2  |
| Serious Eye Damage/Eye Irritation                | Category 2  |
| Germ Cell Mutagenicity                           | Category 1B |
| Carcinogenicity                                  | Category 1B |
| Reproductive Toxicity                            | Category 1B |
| Specific target organ toxicity (single exposure) | Category 3  |
| Target Organs - Respiratory system.              |             |

#### Label Elements

**Signal Word**

Danger

**Hazard Statements**

Combustible liquid  
Causes skin irritation  
Causes serious eye irritation  
Toxic if inhaled  
May cause respiratory irritation  
May cause genetic defects  
May cause cancer  
May damage fertility or the unborn child  
Harmful if swallowed or in contact with skin

**Precautionary Statements****Prevention**

Obtain special instructions before use  
Do not handle until all safety precautions have been read and understood  
Use personal protective equipment as required  
Wash face, hands and any exposed skin thoroughly after handling  
Do not eat, drink or smoke when using this product  
Avoid breathing dust/fume/gas/mist/vapors/spray  
Use only outdoors or in a well-ventilated area  
Wear eye/face protection  
Keep away from heat/sparks/open flames/hot surfaces. - No smoking  
Keep cool

**Response**

IF exposed or concerned: Get medical attention/advice

**Inhalation**

IF INHALED: Remove victim to fresh air and keep at rest in a position comfortable for breathing  
Call a POISON CENTER or doctor/physician

**Skin**

IF ON SKIN: Wash with plenty of soap and water  
Call a POISON CENTER or doctor/physician if you feel unwell  
If skin irritation occurs: Get medical advice/attention  
Take off contaminated clothing and wash before reuse

**Eyes**

IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
If eye irritation persists: Get medical advice/attention

**Ingestion**

IF SWALLOWED: Call a POISON CENTER or doctor/physician if you feel unwell  
Rinse mouth

**Fire**

In case of fire: Use CO<sub>2</sub>, dry chemical, or foam for extinction

**Storage**

Store locked up  
Store in a well-ventilated place. Keep container tightly closed

**Disposal**

Dispose of contents/container to an approved waste disposal plant

**Hazards not otherwise classified (HNOC)**

None identified

### 3. Composition/Information on Ingredients

| Component    | CAS No     | Weight % |
|--------------|------------|----------|
| (S)-Glycidol | 60456-23-7 | 100      |

### 4. First-aid measures

|                                            |                                                                                                                                                                                                                                                                                                                                |
|--------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>General Advice</b>                      | If symptoms persist, call a physician. Show this safety data sheet to the doctor in attendance.                                                                                                                                                                                                                                |
| <b>Eye Contact</b>                         | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Immediate medical attention is required.                                                                                                                                                                                              |
| <b>Skin Contact</b>                        | Wash off immediately with plenty of water for at least 15 minutes. Immediate medical attention is required.                                                                                                                                                                                                                    |
| <b>Inhalation</b>                          | Remove to fresh air. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Immediate medical attention is required. If not breathing, give artificial respiration. |
| <b>Ingestion</b>                           | Do NOT induce vomiting. Call a physician or poison control center immediately.                                                                                                                                                                                                                                                 |
| <b>Most important symptoms and effects</b> | Difficulty in breathing. Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting                                                                                                                                                                                                                   |
| <b>Notes to Physician</b>                  | Treat symptomatically                                                                                                                                                                                                                                                                                                          |

### 5. Fire-fighting measures

|                                         |                                                                                                                                         |
|-----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| <b>Suitable Extinguishing Media</b>     | Water spray, carbon dioxide (CO <sub>2</sub> ), dry chemical, alcohol-resistant foam. Water mist may be used to cool closed containers. |
| <b>Unsuitable Extinguishing Media</b>   | No information available                                                                                                                |
| <b>Flash Point</b>                      | 66 °C / 150.8 °F                                                                                                                        |
| <b>Method -</b>                         | No information available                                                                                                                |
| <b>Autoignition Temperature</b>         | No information available                                                                                                                |
| <b>Explosion Limits</b>                 |                                                                                                                                         |
| <b>Upper</b>                            | No data available                                                                                                                       |
| <b>Lower</b>                            | No data available                                                                                                                       |
| <b>Sensitivity to Mechanical Impact</b> | No information available                                                                                                                |
| <b>Sensitivity to Static Discharge</b>  | No information available                                                                                                                |

#### Specific Hazards Arising from the Chemical

Combustible material. Thermal decomposition can lead to release of irritating gases and vapors. Containers may explode when heated. Keep product and empty container away from heat and sources of ignition.

#### Hazardous Combustion Products

Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>).

#### Protective Equipment and Precautions for Firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

#### NFPA

Health  
3

Flammability  
2

Instability  
0

Physical hazards  
N/A

## 6. Accidental release measures

|                                             |                                                                                                                                                                                                                         |
|---------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Personal Precautions</b>                 | Use personal protective equipment as required. Keep people away from and upwind of spill/leak. Evacuate personnel to safe areas. Remove all sources of ignition. Take precautionary measures against static discharges. |
| <b>Environmental Precautions</b>            | Should not be released into the environment. See Section 12 for additional Ecological Information.                                                                                                                      |
| <b>Methods for Containment and Clean Up</b> | Soak up with inert absorbent material. Keep in suitable, closed containers for disposal. Remove all sources of ignition. Use spark-proof tools and explosion-proof equipment.                                           |

## 7. Handling and storage

|                 |                                                                                                                                                                                                                                                                                                                           |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Handling</b> | Use only under a chemical fume hood. Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Avoid ingestion and inhalation. Keep away from open flames, hot surfaces and sources of ignition. Use only non-sparking tools. Use spark-proof tools and explosion-proof equipment. |
| <b>Storage.</b> | Keep away from heat, sparks and flame. Keep refrigerated. Keep containers tightly closed in a dry, cool and well-ventilated place. Incompatible Materials. Strong oxidizing agents.                                                                                                                                       |

## 8. Exposure controls / personal protection

|                                             |                                                                                                                                                                                                                                                       |
|---------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b><u>Exposure Guidelines</u></b>           | This product does not contain any hazardous materials with occupational exposure limitsestablished by the region specific regulatory bodies.                                                                                                          |
| <b>Engineering Measures</b>                 | Use only under a chemical fume hood. Ensure that eyewash stations and safety showers are close to the workstation location. Use explosion-proof electrical/ventilating/lighting equipment. Ensure adequate ventilation, especially in confined areas. |
| <b><u>Personal Protective Equipment</u></b> |                                                                                                                                                                                                                                                       |
| <b>Eye/face Protection</b>                  | Wear appropriate protective eyeglasses or chemical safety goggles as described by OSHA's eye and face protection regulations in 29 CFR 1910.133 or European Standard EN166.                                                                           |
| <b>Skin and body protection</b>             | Wear appropriate protective gloves and clothing to prevent skin exposure.                                                                                                                                                                             |
| <b>Respiratory Protection</b>               | Follow the OSHA respirator regulations found in 29 CFR 1910.134 or European Standard EN 149. Use a NIOSH/MSHA or European Standard EN 149 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.     |
| <b>Recommended Filter type:</b>             | Organic gases and vapours filter. Type A. Brown. conforming to EN14387.                                                                                                                                                                               |
| <b>Hygiene Measures</b>                     | When using do not eat, drink or smoke. Provide regular cleaning of equipment, work area and clothing.                                                                                                                                                 |

## 9. Physical and chemical properties

|                            |                               |
|----------------------------|-------------------------------|
| <b>Physical State</b>      | Liquid                        |
| <b>Appearance</b>          | Light yellow                  |
| <b>Odor</b>                | No information available      |
| <b>Odor Threshold</b>      | No information available      |
| <b>pH</b>                  | No information available      |
| <b>Melting Point/Range</b> | -54 °C / -65.2 °F             |
| <b>Boiling Point/Range</b> | 160 - 161 °C / 320 - 321.8 °F |
| <b>Flash Point</b>         | 66 °C / 150.8 °F              |
| <b>Evaporation Rate</b>    | No information available      |

|                                        |                          |
|----------------------------------------|--------------------------|
| Flammability (solid,gas)               | Not applicable           |
| Flammability or explosive limits       |                          |
| Upper                                  | No data available        |
| Lower                                  | No data available        |
| Vapor Pressure                         | 1 mmHg @ 20°C            |
| Vapor Density                          | No information available |
| Specific Gravity                       | 1.116                    |
| Solubility                             | No information available |
| Partition coefficient; n-octanol/water | No data available        |
| Autoignition Temperature               | No information available |
| Decomposition Temperature              | No information available |
| Viscosity                              | No information available |
| Molecular Formula                      | C3 H6 O2                 |
| Molecular Weight                       | 74.08                    |

## 10. Stability and reactivity

|                                  |                                                                                                       |
|----------------------------------|-------------------------------------------------------------------------------------------------------|
| Reactive Hazard                  | None known, based on information available                                                            |
| Stability                        | Stable under normal conditions.                                                                       |
| Conditions to Avoid              | Incompatible products. Excess heat. Keep away from open flames, hot surfaces and sources of ignition. |
| Incompatible Materials           | Strong oxidizing agents                                                                               |
| Hazardous Decomposition Products | Carbon monoxide (CO), Carbon dioxide (CO <sub>2</sub> )                                               |
| Hazardous Polymerization         | Hazardous polymerization does not occur.                                                              |
| Hazardous Reactions              | None under normal processing.                                                                         |

## 11. Toxicological information

### Acute Toxicity

#### Product Information

#### Component Information

|                                      |                          |
|--------------------------------------|--------------------------|
| Toxicologically Synergistic Products | No information available |
|--------------------------------------|--------------------------|

#### Delayed and immediate effects as well as chronic effects from short and long-term exposure

|                 |                                                 |
|-----------------|-------------------------------------------------|
| Irritation      | Irritating to eyes, respiratory system and skin |
| Sensitization   | No information available                        |
| Carcinogenicity | May cause cancer.                               |

| Component    | CAS No     | IARC       | NTP        | ACGIH      | OSHA       | Mexico     |
|--------------|------------|------------|------------|------------|------------|------------|
| (S)-Glycidol | 60456-23-7 | Not listed | Not listed | Not listed | Not listed | Not listed |

|                   |                                       |
|-------------------|---------------------------------------|
| Mutagenic Effects | Possible risk of irreversible effects |
|-------------------|---------------------------------------|

|                          |                           |
|--------------------------|---------------------------|
| Reproductive Effects     | May impair fertility.     |
| Developmental Effects    | No information available. |
| Teratogenicity           | No information available. |
| STOT - single exposure   | Respiratory system        |
| STOT - repeated exposure | None known                |

|                                                   |                                                                                     |
|---------------------------------------------------|-------------------------------------------------------------------------------------|
| <b>Aspiration hazard</b>                          | No information available                                                            |
| <b>Symptoms / effects, both acute and delayed</b> | Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting |
| <b>Endocrine Disruptor Information</b>            | No information available                                                            |
| <b>Other Adverse Effects</b>                      | The toxicological properties have not been fully investigated.                      |

## 12. Ecological information

### Ecotoxicity

Do not empty into drains.

|                                      |                           |
|--------------------------------------|---------------------------|
| <b>Persistence and Degradability</b> | No information available  |
| <b>Bioaccumulation/ Accumulation</b> | No information available. |
| <b>Mobility</b>                      | No information available. |

## 13. Disposal considerations

|                               |                                                                                                                                                                                                                                                                 |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Waste Disposal Methods</b> | Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification. |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## 14. Transport information

### DOT

|                             |                               |
|-----------------------------|-------------------------------|
| <b>UN-No</b>                | UN2810                        |
| <b>Proper Shipping Name</b> | Toxic liquid, organic, n.o.s. |
| <b>Technical Name</b>       | (S)-Glycidol                  |
| <b>Hazard Class</b>         | 6.1                           |
| <b>Packing Group</b>        | III                           |

### TDG

|                             |                               |
|-----------------------------|-------------------------------|
| <b>UN-No</b>                | UN2810                        |
| <b>Proper Shipping Name</b> | Toxic liquid, organic, n.o.s. |
| <b>Hazard Class</b>         | 6.1                           |
| <b>Packing Group</b>        | III                           |

### IATA

|                             |                               |
|-----------------------------|-------------------------------|
| <b>UN-No</b>                | UN2810                        |
| <b>Proper Shipping Name</b> | Toxic liquid, organic, n.o.s. |
| <b>Hazard Class</b>         | 6.1                           |
| <b>Packing Group</b>        | III                           |

### IMDG/IMO

|                             |                               |
|-----------------------------|-------------------------------|
| <b>UN-No</b>                | UN2810                        |
| <b>Proper Shipping Name</b> | Toxic liquid, organic, n.o.s. |
| <b>Hazard Class</b>         | 6.1                           |
| <b>Packing Group</b>        | III                           |

## 15. Regulatory information

### United States of America Inventory

| Component    | CAS No     | TSCA | TSCA Inventory notification - Active-Inactive | TSCA - EPA Regulatory Flags |
|--------------|------------|------|-----------------------------------------------|-----------------------------|
| (S)-Glycidol | 60456-23-7 | -    | -                                             | -                           |

### Legend:

**TSCA** US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

X - Listed

'-' - Not Listed

**TSCA - Per 40 CFR 751, Regulation of Certain Chemical Substances & Mixtures, Under TSCA Section 6(h) (PBT)**

Not applicable

**TSCA 12(b)** - Notices of Export

Not applicable

**International Inventories**

Canada (DSL/NDSL), Europe (EINECS/ELINCS/NLP), Philippines (PICCS), Japan (ENCS), Japan (ISHL), Australia (AICS), China (IECSC), Korea (KECL).

| Component    | CAS No     | DSL | NDSL | EINECS | PICCS | ENCS | ISHL | AICS | IECSC | KECL |
|--------------|------------|-----|------|--------|-------|------|------|------|-------|------|
| (S)-Glycidol | 60456-23-7 | -   | -    | -      | -     | -    | -    | -    | -     | -    |

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

**U.S. Federal Regulations****SARA 313**

Section 313 of Title III of the Superfund Amendments and Reauthorization Act of 1986 (SARA). This product does not contain any chemicals which are subject to the reporting requirements of the Act and Title 40 of the Code of Federal Regulations, Part 372

**SARA 311/312 Hazard Categories**

Should this product meet EPCRA 311/312 Tier reporting criteria at 40 CFR 370, refer to Section 2 of this SDS for appropriate classifications.

**CWA (Clean Water Act)**

Not applicable

**Clean Air Act**

Not applicable

**OSHA - Occupational Safety and Health Administration**

Not applicable

**CERCLA**

This material, as supplied, does not contain any substances regulated as hazardous substances under the Comprehensive Environmental Response Compensation and Liability Act (CERCLA) (40 CFR 302) or the Superfund Amendments and Reauthorization Act (SARA) (40 CFR 355). There may be specific reporting requirements at the local, regional, or state level pertaining to releases of this material.

**California Proposition 65**

This product does not contain any Proposition 65 chemicals.

**U.S. State Right-to-Know Regulations**

Not applicable

**U.S. Department of Transportation**

Reportable Quantity (RQ): N  
DOT Marine Pollutant N  
DOT Severe Marine Pollutant N

**U.S. Department of Homeland Security**

This product does not contain any DHS chemicals.

**Other International Regulations****Mexico - Grade**

Moderate risk, Grade 2

## Authorisation/Restrictions according to EU REACH

Not applicable

| Component    | CAS No     | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|--------------|------------|---------------------------------------------------------------------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| (S)-Glycidol | 60456-23-7 | -                                                                   | -                                                                             | -                                                                                                     |

## Safety, health and environmental regulations/legislation specific for the substance or mixture

| Component    | CAS No     | OECD HPV       | Persistent Organic Pollutant | Ozone Depletion Potential | Restriction of Hazardous Substances (RoHS) |
|--------------|------------|----------------|------------------------------|---------------------------|--------------------------------------------|
| (S)-Glycidol | 60456-23-7 | Not applicable | Not applicable               | Not applicable            | Not applicable                             |

## Contains component(s) that meet a 'definition' of per &amp; poly fluoroalkyl substance (PFAS)?

Not applicable

## Other International Regulations

| Component    | CAS No     | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|--------------|------------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------|------------------------------------|
| (S)-Glycidol | 60456-23-7 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |

## 16. Other information

## Prepared By

Health, Safety and Environmental Department  
 Email: chem.techinfo@thermofisher.com  
 www.thermofisher.com

## Creation Date

16-Jan-2009

## Revision Date

26-Mar-2024

## Print Date

26-Mar-2024

## Revision Summary

New emergency telephone response service provider.

## Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

End of SDS